

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Third Semester B.Tech Degree Regular and Supplementary Examination December 2022
(2019 Scheme)

Course Code: EST200

Course Name: Design and Engineering

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all questions. Each question carries 3 marks Marks

1. List the objectives and constraints for designing a school bag for school students.

Objectives

1. Light weight
2. Water resistant, durable material
3. Minimum strain for the shoulders (comfortable to carry)

Constraints

1. Maximum weight to be limited to 250 g
2. Cost factor

2. State how engineering design is different from other kinds of design.

Engineering design is a thoughtful process for generating plans or schemes for devices, systems, or processes that attain given objectives while adhering to specified constraints.

The phases of design process are

- Identifying and detailing Customer Requirements
- Setting Design Objectives
- Identifying Constraints
- Establishing Functions
- Generating Design Alternatives
- Choosing the Best Feasible Design

3. How to perform design thinking as a team managing the conflicts?

Design thinking is one of those methodologies useful for more than designing, more than thinking, and more than innovation.

The probable steps to be followed are

Step 1: Thoroughly understanding the problem

Step 2: Thinking of potential alternatives (solutions)

Step 3: Experimenting

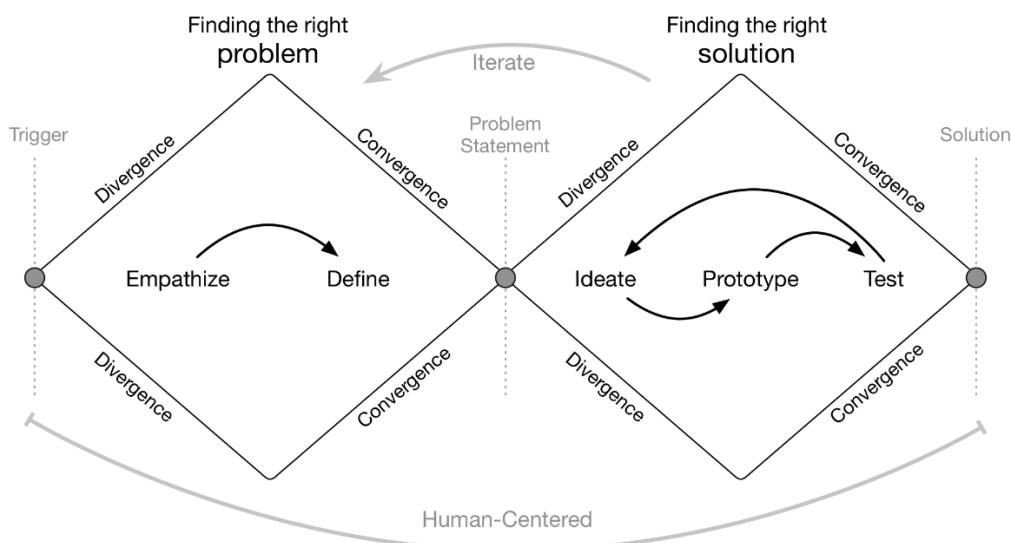
Step 4: Scaling or Pivoting

4. Explain divergent-convergent questioning in design thinking.

Divergence and convergence are like the inhale and exhale of the design thinking process. Divergence creates space and possibilities, while convergence brings focus and direction.

Divergence is the process of exploring and generating a wide range of ideas and possibilities without any judgment or criticism. Divergent thinking allows designers to expand the possibilities and consider different perspectives, angles, and dimensions of the problem.

Convergent thinking refers to the process of selecting, evaluating, and refining the most promising ideas from the divergent phase. It involves narrowing down and refining possibilities by selecting the best ideas and combining them to create a coherent and effective solution.



5. List the factors to be considered in preparing oral presentations to communicate.

Because of the variety of presentations and briefings that a team may be called upon to make, it is impossible to examine each of them in detail. However, there are key elements common to most of them.

Key element of oral presentations are

1. Identify the audience
2. Outline the presentation
3. Develop appropriate supporting materials
4. Practice presentation.

6. How does prototyping help to predict whether the design will function well or not?

A prototype is defined as an initial or preliminary version from which other forms are developed. This provides you with insight into the functionality of your design and any changes needed in order to make your work a pleasure to use.

In this context, prototypes are working models of designed artifacts. They are tested in the same operating environments in which they're expected to function as final products.

Prototyping can benefit designs in following ways

- Exposing the key areas of improvement in the design
- Helping to iterate quickly on design concepts
- Making It easy for non-technical clients to understand
- Reducing development time and cost
- Helps to choose the suitable variant
- Testing the responsiveness of a design
- Creating a robust feedback circle
- Keep the focus\

7. Relate how designs are inspired from nature.

Biomimicry (or biomimetics) is what guides Nature inspired design, representing the bridge between biology and design principles.

Biomimicry is an approach to innovation that seeks sustainable solutions to human challenges by emulating nature's time-tested patterns and strategies. In short, biomimicry is the process of taking the innovations that exist in nature and applying them to technology.

Some examples are

- Shinkansen Bullet Train by Japan- inspired from Kingfisher
- Sharkskin Paint by IFAM - inspired from skin of shark
- Butterfly wing scale photonics by the University of Cambridge's NanoPhotonics Centre - structure can encrypt information in optical signatures on banknotes, passports, or other valuable items to prevent forgery.
- Velcro - inspired from simple design of tiny hooks at the end of the burr's spines
- Lotus-inspired hydrophobia

8. How does the life cycle design approach influence design decisions?

Life-Cycle Design is the environmentally sound design of products based on the whole lifecycle starting from exploitation and processing of raw materials, pre production, production, distribution, to use and returning materials back into the industrial cycles

9. What are design rights and how can an engineer put it into practice?

The design right gives you the exclusive right to the commercial use of a new and distinctive design for a certain period of time. This exclusive right is also known as design protection. This allows you to prohibit others from using the design without your permission.

10. Describe the role of ethics in the design of any two products.

These days usability should be a basic requirement. An unusable product is considered a design failure. More specifically the design should help the user accomplish what they want, meet their needs, and be easy and pleasant to use.

1. Products are always designed for the “targeted customer” but think of who is (un)intentionally left out. Often these are folks with disabilities. As an example, website design is not always optimized for those with vision impairment despite the fact that, according to the World Health Organization, at least 1 billion people are blind or visually impaired.

2. The best ethical design practice would be to develop designs that only collect personal information that is in the best interest of the users. For example, Signal is a secure phone and messenger app specifically designed to protect its user’s privacy.

PART B

Answer any one full question from each module. Each question carries 14 marks

Module 1

11. Show the design of a length adjustable mop to clean the ceiling fan. Use hand sketches to illustrate the various stages of the design process.

Problem: Show the designing of a length adjustable mop to clean ceiling fan

1. Identifying Customer Needs :

Most of the time the customer provides a generic statement of need, and it is up to the engineer to identify the specific needs of the customer.

- Portable
- Easy handling
- Adjustable height
- Smooth operation
- Low weight

2. Defining Goals (Setting design Objective)

The definition is a general statement of the desired end product.

Many of the difficulties encountered in design may be traced to poorly stated goals, or goals that were hastily written and resulted in confusion or too much flexibility. Designers must recognize that customer needs are not the same as product specifications

- Flexible Cleaning
- The Rod Material is made from stainless/ plastic steel, hard to break, and long service life.
- Light weight
- Height can be adjusted by the help of control
- Washable material
- Multi purpose use-You can use this mop to clean the interior roof, high ceiling, ceiling fan, blinds, cobwebs, gap dust, car, furniture, and lamps, etc.
- reduces the human effort in case of large

3. Identifying design constraints

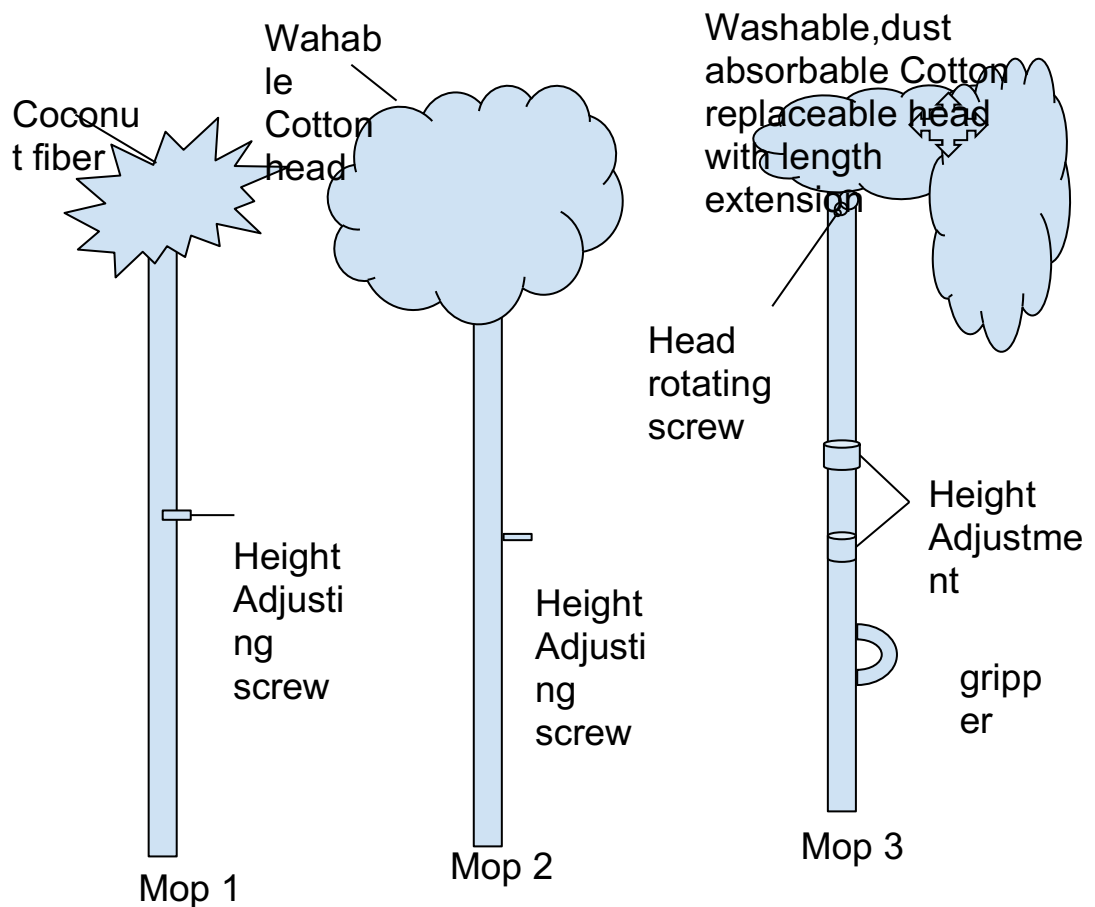
- Cost of the mop should less than 400 Rs
- Should resist corrosion

- Need flexible cleaning in every direction
- Height should be adjustable so that multi users can use the mop for cleaning the ceiling fan.
- Adjustable Handle
- The head material should be eco friendly, waterproof and replaceable.

4. Establishing functions.

- The head can bend at multiple angles for dusting your room in an effective way; Can be applied to dust ceiling fans, window sills, walls, corners, and other places difficult to reach.
- Long Handle Allows You To Reach Hard-To-Reach Corners,
- The head is Soft, Super-Absorbent and Quick Drying Duster
- It should be simple and lightweight.

5. Generate design alternatives



MOP1

Pros:

It's height adjustable. It's made up of a plastic rod and coconut fiber head. It is low cost.

Cons :

The rod of the mop breakable, the coconut fiber head doesn't absorb dust and water.

MOP2

Pros:

It's height adjustable. It's made up of a plastic rod and washable cotton head. It is low cost.

Cons :

The cotton head cannot be rotated.

MOP3

Pros:

It's height adjustable. It's made up of a stainless steel rod and it cannot be easily breakable. It has a washable expandable rotatable cotton head. Using the shaft the head can be rotated in any direction. It is provided with a gripper for comfort. The head can be replaced easily.

Cons :

The cost of this mop is higher than the others.

6. Choosing the best feasible design

From the three design mop 3 it is best suitable for cleaning. It can be used by multiple users. The head can be replaced easily. This is a long lasting product as compared to others.

12. In certain situations, users require extra length for their mobile charger cable. But offering extra cable length becomes an issue while normal usage. Develop a design to effectively solve this problem. Use necessary hand sketches. Also give the objectives and constraints of your design.

Problem: Variable length mobile charger

1. Identifying Customer Needs :

Most of the time the customer provides a generic statement of need, and it is up to the engineer to identify the specific needs of the customer.

- Need variable length mobile charger
- Fast charging
- charger with multiple output ports
- Chargers with constant charging voltages
-

2. Defining Goals (Setting design Objective)

The definition is a general statement of the desired end product.

Many of the difficulties encountered in design may be traced to poorly stated goals, or goals that were hastily written and resulted in confusion or too much flexibility. Designers must recognize that customer needs are not the same as product specifications

- Need variable length mobile charger
- Fast charging
- Chargers with constant charging voltages
- Smart charger
- Chargers with high quality components and materials
- Charger protection

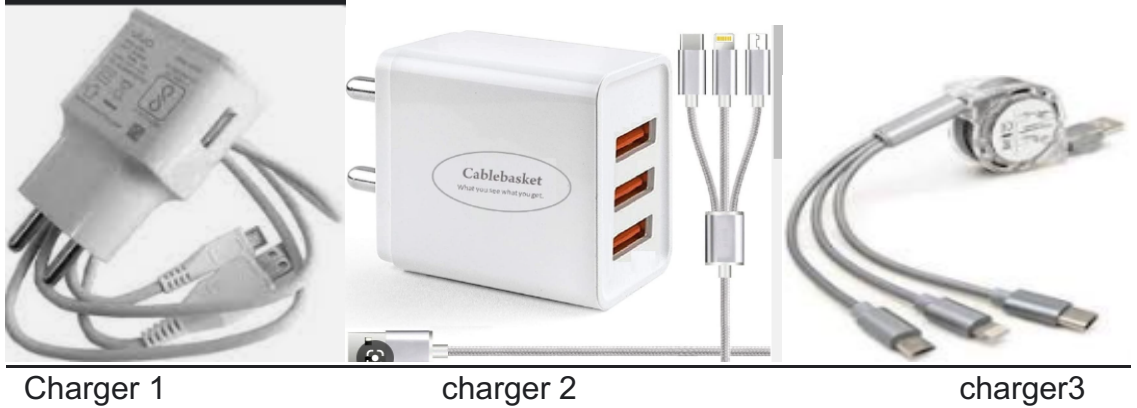
3. Identifying design constraints

- Cost of the charger should be less than 500 Rs
- length can vary from cm to meters
-

4. Establishing functions.

- Adjustable length Wall Charger
- Suitable For: Mobile
- Universal Voltage
- Output Current : 3.1 A
- 20 Watt PD Usb C fast-charge adapter, for the fastest charging output
- Multi charging facility

5. Generate design alternatives



6. Choosing the best feasible design

From the three models the length of charger 3 can be varied and can be used for the multiple types of chargers. So it is the best choice for charging.

Module 2

13. Design a suitable product for easy cleaning of dust from the chalkboards in a (14) classroom. Illustrate the various stages involved in design thinking. Sketch the final design.

Design of Blackboard Duster Cleaning Machine

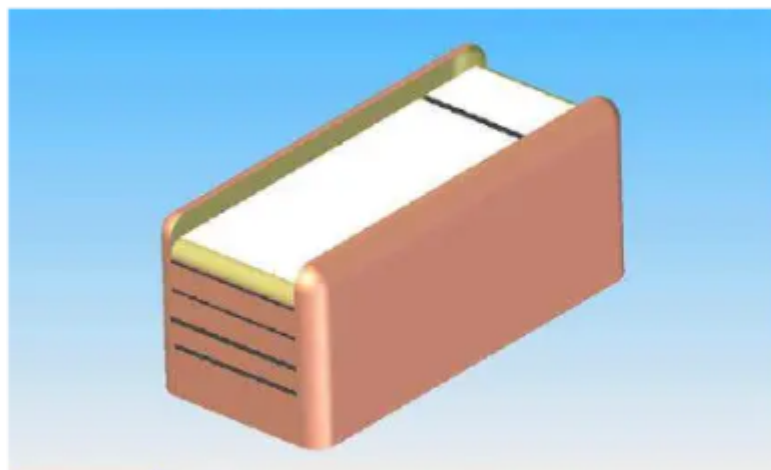


Figure 1: Model of a Duster Cleaner using Vacuum Pump.

14. Construct two possible designs of a dining table set that occupies minimum space

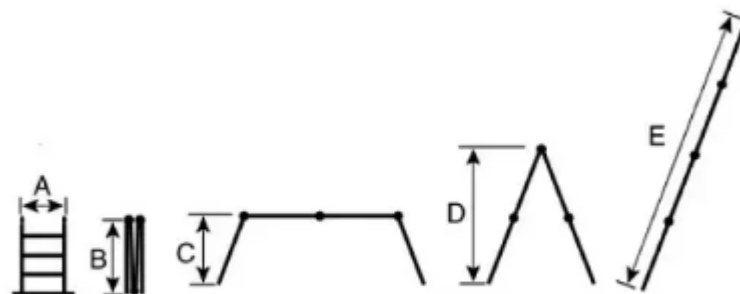
when not in use and then refine them to narrow down to the best design. Show how divergent-convergent thinking helps in the process. Provide your rationale for each step by using hand sketches only.





Module 3

15. Graphically communicate the design of a foldable ladder for electricians. Draw (14) the detailed 2D drawings of the same with design detailing, material selection, scale drawings and dimensions. Use only hand sketches.



Material Selection for ladder:

Ladders can be made of metal, wood, reinforced plastic or fiberglass. The most common material is aluminum. Metal ladders should never be used for work on or around exposed electrical elements; a wood or fiberglass ladder is necessary.

16. Describe the role of mathematical modeling in design engineering. Show how mathematics and physics play a role in designing a simple paper cutting scissor.

Fundamental and quantitative way to understand and analyze complex systems and phenomena

Complement to Theory and Experiments, and often Intergate them

Is often used in place of experiments when experiments are too large, too expensive, too dangerous, or too time consuming.

Can be useful in “what if” studies; e.g. to investigate the use of pathogens (viruses, bacteria) to control an insect population.

Is a modern tool for scientific investigation.

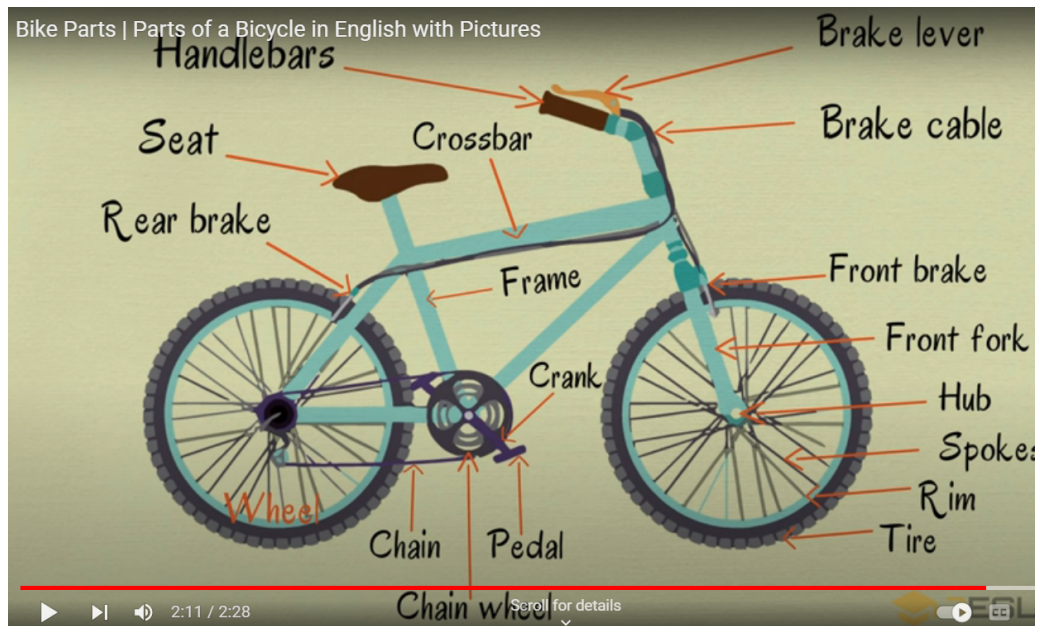
Math and physics are very much part of the design process.

Physics Behind Scissor: In a pair of scissors, the fulcrum is present in between the load (the cutting blades) and the effort (the handles) which make them a class I lever.

When you cut an object with a pair of scissors, the force that you feel between your fingers includes two main components: friction forces of the contact of the blades, and the forces of cutting of the object

Module 4

17. Design a nature inspired solar lamp for the students residing in urban areas. These students do not have proper availability of electricity and cannot afford highly priced products. Illustrate your design with sketches.
18. Apply modular engineering to a conventional bicycle and design a bicycle which can be used in different terrains. Illustrate the design using sketches.



Today bike manufacturers often create designs to support a specific terrain type. The most obvious example is the mountain bike.

Finding The Right Bike for the Right Terrain



Road Bikes

Has drop or turned-down handlebars, narrow tires, and has lightweight frame. Ideal for a variety of pavement applications.



Mountain Bikes

Have unique features that handle a steeper terrain, which is why they often have lower gears than most other road cycles.

Module 5

19. Examine the changes in the design of a footwear in terms of production methods, reliability issues and environmental factors. Use hand sketches and give proper rationalization for the changes in design.

Design of a foot wear

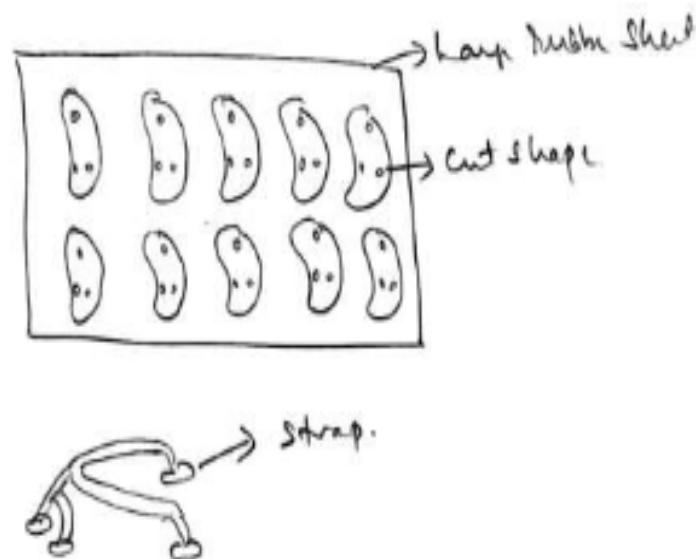
A design of foot wear is simple. We have to consider certain simple things while we are designing with the given constraints

Let us discuss each

Constraints of production method

Since we are designing a rubber foot wear we can have a good rubber material for foot wear with long lasting life

- Foot wear is produced in bulk so we need large amount of production materials
- So, to reduce the cost we can use large sheets of rubber material and cut the foot wear from that using machines
- The other works like stitching or combining parts of the foot wear are being done with workers so labours are required to do those jobs
- For stitching we can use good quality nylons



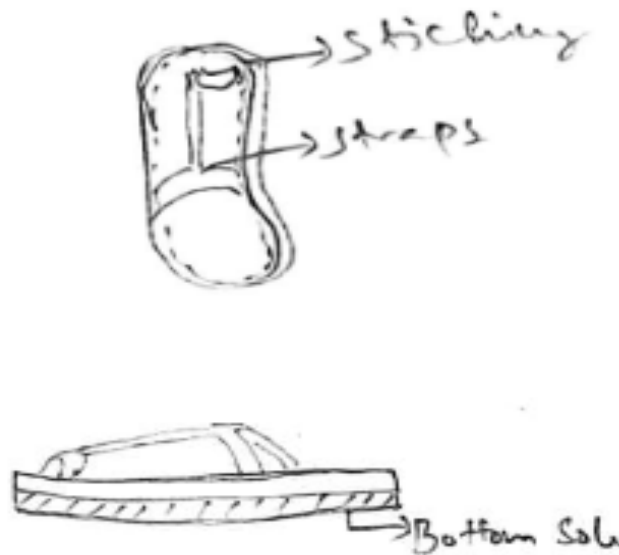
Constraints of life span requirements

- By considering life span requirement of a foot wear it is difficult to consider the return of the product and recycling. So we must make the product with a bio degradable material. The best option for that is good quality rubber

- Another problem may be manufacturing defects or manmade error, We can provide a service for the foot wear around 1-2 year for that
- On the life span another issue is using the foot wear at the time of rainy seasons. By using the rubber product for the manufacturing, we can solve that issue
- To get the old product back and recycle. we can give some discounts for the old product for new purchases. By giving that way we can make sure the return of the product after the use and done for the recycling purpose

Constraints of reliability

- One of the main reliability issues is long life of the foot wear on all seasons. This constraints can be resolved using the good footwear material rubber
- Another constraints is breaking the straps. This can be reduced by strong materials at the end and also by stitching the joints of the foot wear after the instalment of the strap
- Wear and tear at the corners of the foot wear can be also reduced by stitching around the corners of the foot wear with good nylon.
- Another constrain is wear the bottom side of the foot wear while at the long use. To increas the life of the foot wear we can use strong and bendable quality rubber material at the bottom of the foot wear as souls of the foot wear



Constraints of the environmental factors

- The product like foot wear the environmental factors are mainly use of the product at the time of rainy season. As we discussed before, making of the product with good quality factors can be extend the life time of the foot wear
- Due to govt. orders the product used for the making of foot wear must be bio degradable. We already mentioned that in the previous steps
- Since we are also considered the recycling steps and initiation for the products. It will help more in the environmental factors.
- We must also follow the govt. rules for the machine used in the manufacturing factory. Or we can import the product from other merchandises which will solve the other pollution problems and cost.

Natural rubber, cork, bamboo, and organic cotton are a few examples of eco-friendly and renewable natural materials can be incorporated during shoe manufacturing.

20. Describe how to estimate the cost of a residence house in the design stage. Show how economics will influence its design. Use hand sketches to support your arguments.

Mainly for a product design there are three costs 1) labour cost , 2) material cost and 3) overhead cost.
